



JOB DATA *Pipeline*

TEAM A

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The Problem

- ## Raw job data cannot be directly used for reliable analysis.

Build a repeatable Data Engineering pipeline that transforms fragmented job postings into a standardized, reliable, and analysis-ready dataset.

Identify & evaluate suitable job data sources.

Collect & explore job posting data.

Clean & standardize job posting information.

Handle data quality issues including missing, duplicate, and outdated records.

Build a repeatable Data Engineering pipeline.

Produce a curated, reliable, and analysis-ready dataset.

PROJECT OBJECTIVE



WHY SAUDI ARABIA AS OUR TARGET MARKET ?

Target Market



Rapidly evolving
labor market.

Multiple digital
employment
platforms.

High demand for
job-market
insights.

Opportunity to build
a localized Saudi
job dataset.

Potential
DATA
SOURCES



Potential DATA SOURCES



Source	Platform	Access	HTTP	Why consider it?
✓ Greenhouse ✓ Lever ✓ Ashby ✓ Workable ✓ SmartRecruiters ✓ Recrutee	ATS	Public API	GET	Public REST API by design, no auth, structured JSON response. Confirmed strong Saudi job coverage across multiple boards.
✓ Jooble	Job aggregator	Authenticated API	POST	Cross-platform aggregator, Saudi-region API key. Confirmed very high volume and precision of Saudi job listings.

Potential DATA SOURCES

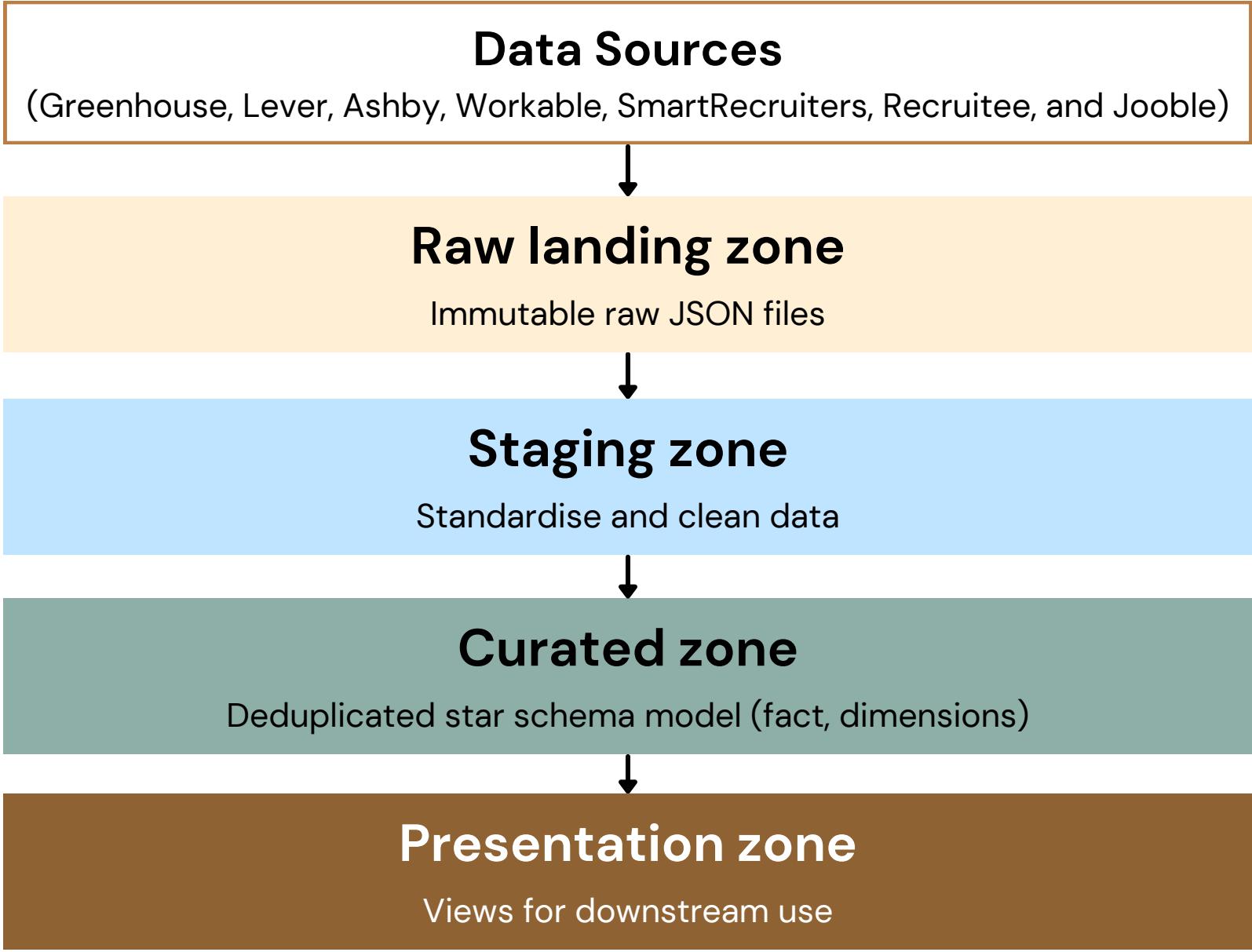


Source	Reason
✗ LinkedIn ✗ Indeed ✗ GulfTalent ✗ NaukriGulf ✗ Glassdoor ✗ Bayt	Terms of use and robots.txt explicitly prohibit automated data collection.
✗ Jadarat ✗ Taqat ✗ Qiwa	Require authentication through Nafath (Saudi national SSO) — technically and legally inaccessible.

Proposed DIDELINI



DATA FLOW DIAGRAM



MECHANICS & TRANSFORMATION PROTOCOLS

A. Data & Record Validation Rules

- File-Level Check
- Required Fields Check

B. Standardization Logic

- Geography Hierarchy
- Role Normalization
- Employment Type Normalization
- Seniority Normalization
- Work Arrangement Normalization
- Salary Period Standardization (SAR only)

C. Deduplication & Entity Resolution

- Posting Deduplication (Blocking + Semantic Matching + Threshold + Error Cases)
- Employer Entity Resolution

D. Skill & Technology Extraction

E. Idempotency & Repeatability

- SHA-256 posting_business_key
- MERGE-based loading
- Watermark advances only on success
- Staging zone rebuildable from raw

F. Lifecycle & Freshness Tracking

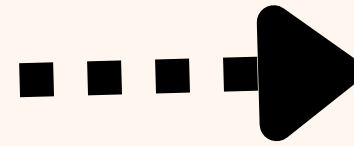
- first_seen_date, last_seen_date,
consecutive misses, disappeared date



Risks & Mitigation

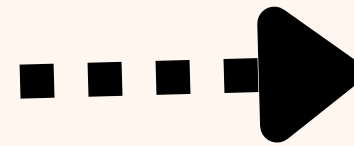


Limited API Access



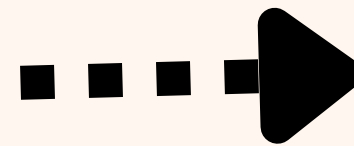
Verify each potential platform for a documented or unofficial API before starting to scrape. ✓

Anti-Scraping



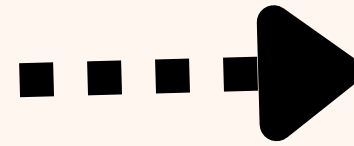
Implement polite scraping practices, including delay/throttling and user-agent rotation. ✓

Incomplete Fields



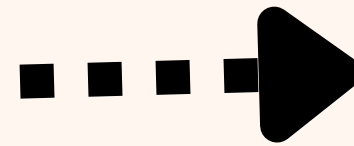
Implement standardized default values like "Unspecified" and include programmatic null checks. ✓

Layout Changes



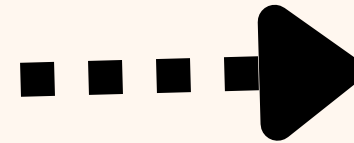
Separate data landing from parsing. Add defensive parsing with schema validation to spot layout changes early, avoiding complete pipeline failures. ✓

Source Feasibility



Retain a broad pool of candidate platforms during testing and Then narrow selection based on empirical evaluation of data ✓

Duplicate Listings



semantic matching strategy that converts text into a numerical representation that captures its meaning. ✓



ROADMAP & DELIVERABLES



Week	Phase	Task	Goal
week 3	Data Ingestion	Collect job data from different sources	Raw data collected
week 4	Cleaning & Deduplication	Clean data and remove duplicates	Clean dataset
week 4	Schema Design	Identify and standardize data fields	Final schema
week 5	Quality Testing	Check data accuracy and completeness	Validated dataset
week 6	Final Delivery	Store and prepare the final dataset	Final job dataset

EXPECTED OUTCOMES

- ✓ **One Job Database:** Collect job postings from different sources in one place.
- ✓ **Useful Job Market Insights:** Helping people looking for jobs understand the required skills and find suitable job opportunities.
- ✓ **Job Market Analysis:** Organize job data, it can be analyzed and used in dashboards.



THANK *You*

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